

COMP5349– Cloud Computing

Week 13: Review and Exam Info

Dr. Ying Zhou

The University of Sydney

Table of Contents

01 Course Review

02 Exam Information

COMMONWEALTH OF AUSTRALIA

Copyright Regulations 1969

WARNING

This material has been reproduced and communicated to you by or on behalf of the **University of Sydney** pursuant to Part VB of the Copyright Act 1968 (the Act).

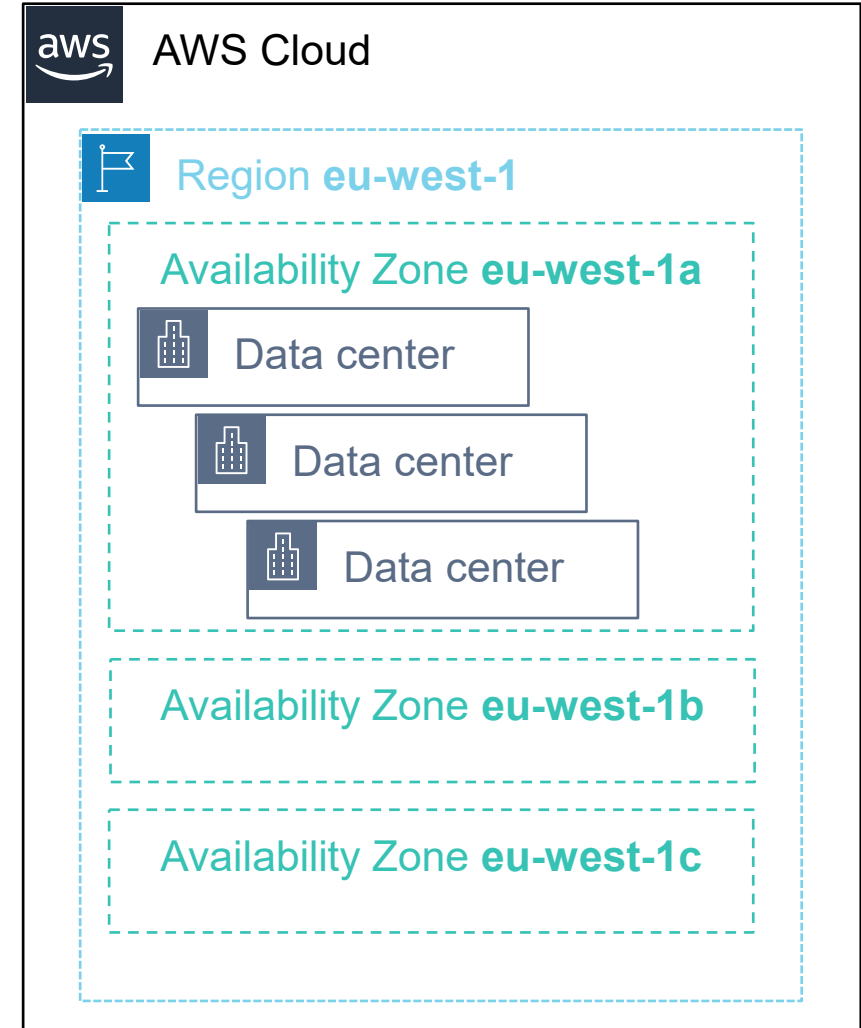
The material in this communication may be subject to copyright under the Act. Any further reproduction or communication of this material by you may be the subject of copyright protection under the Act.

Do not remove this notice

Course Review

AWS Global Infrastructure

- An **AWS Region** is a geographical area.
- A Region typically consists of two or more **Availability Zones**.
- Each **Availability Zone** is a fully isolated partition of the AWS infrastructure.
 - Availability Zones consist of discrete **data centers**
- A data center typically has 50,000 to 80,000 physical servers.



AWS foundational services

Applications



Virtual desktops



Collaboration and sharing

Platform Services

Databases

Relational

NoSQL

Caching

Analytics

computing

Real-time

Data
warehouse

Data
workflows

Application services

Queuing

Orchestration

App Streaming

Transcoding

Email

Search

Deployment and management

Containers

DevOps tools

Resource templates

Usage tracking

Monitoring and logs

Mobile Services

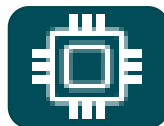
Identity

Sync

Mobile
Analytics

Notifications

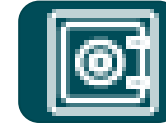
Foundation Services



Compute (virtual, automatic scaling, and load balancing)



Networking



Storage (object, block, and archive)

Infrastructure

Regions

Availability Zones



Edge locations

AWS Global Services

- A few AWS Services operate across the all regions, you don't need to specify region, or az when create or access them
 - **IAM**
 - Route53
 - CloudFront

AWS regional service

- Some AWS Services operates at the region level
 - Resources are deployed at a particular region
- Examples:
 - S3
 - VPC
 - RDS (multi-AZ)
 - CloudFormation
 - Secretes Manager
 - ALB, ASG
 - SQS
 - SNS

AWS Zonal services

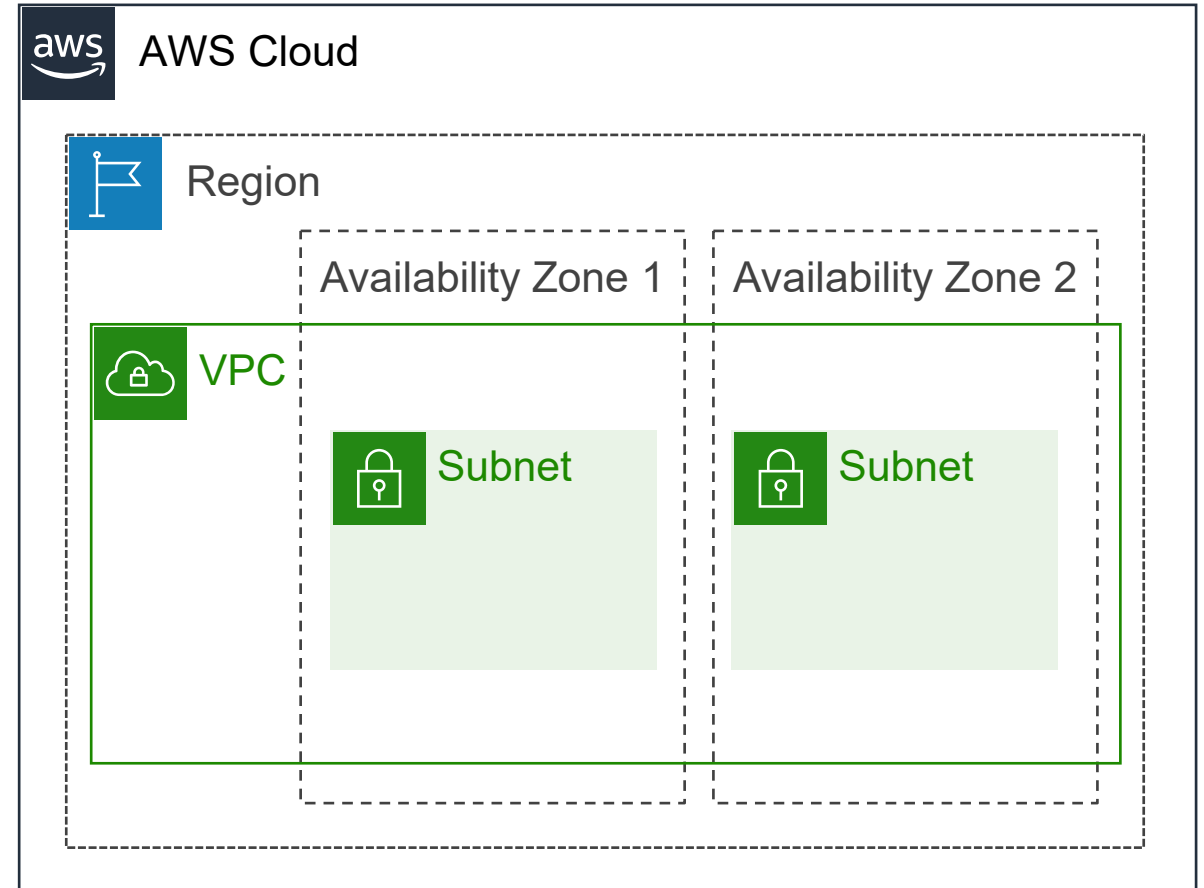
- Some services operate at AZ level
 - You must specify the AZ or equivalent when deploy them
- Examples
 - EC2
 - EBS
 - Single AZ RDS
 - Subnet

Network isolation and access control

- Some services requires users to control network access, may need IP address and security management. Usually those services are placed ***inside*** a VPC
 - EC2, databases and all related services like ALB, ASG, etc
- Other services are fully managed by AWS and does not require user to manage the network access are placed ***outside*** VPC
 - S3, SNS, SQS, DynamoDB, Secretes Manager, IAM, CloudFormation

Amazon Virtual Private Cloud

- VPCs:
 - **Logically isolated** from other VPCs
 - **Dedicated** to an AWS account
 - Belong to a single **AWS Region** and can span multiple Availability Zone
 - Each VPC has a CIDR range
 - Different VPCs can have overlapping CIDR range
- Subnets:
 - **Range of IP addresses** that divide a VPC
 - Belong to a single **Availability Zone**
 - Classified as **public** or **private**



IP addressing

- When you create a VPC, you assign it to an IPv4 **CIDR block** (range of **private** IPv4 addresses).
- You **cannot change the address range** after you create the VPC.
- The **largest** IPv4 CIDR block size is **/16**.
- The **smallest** IPv4 CIDR block size is **/28**.
- IPv6 is also supported (with a different block size limit).
- CIDR blocks of subnets **cannot overlap**.

 VPC

16 bits for host address

x.x.x.x/16 or 65,536 addresses (max)

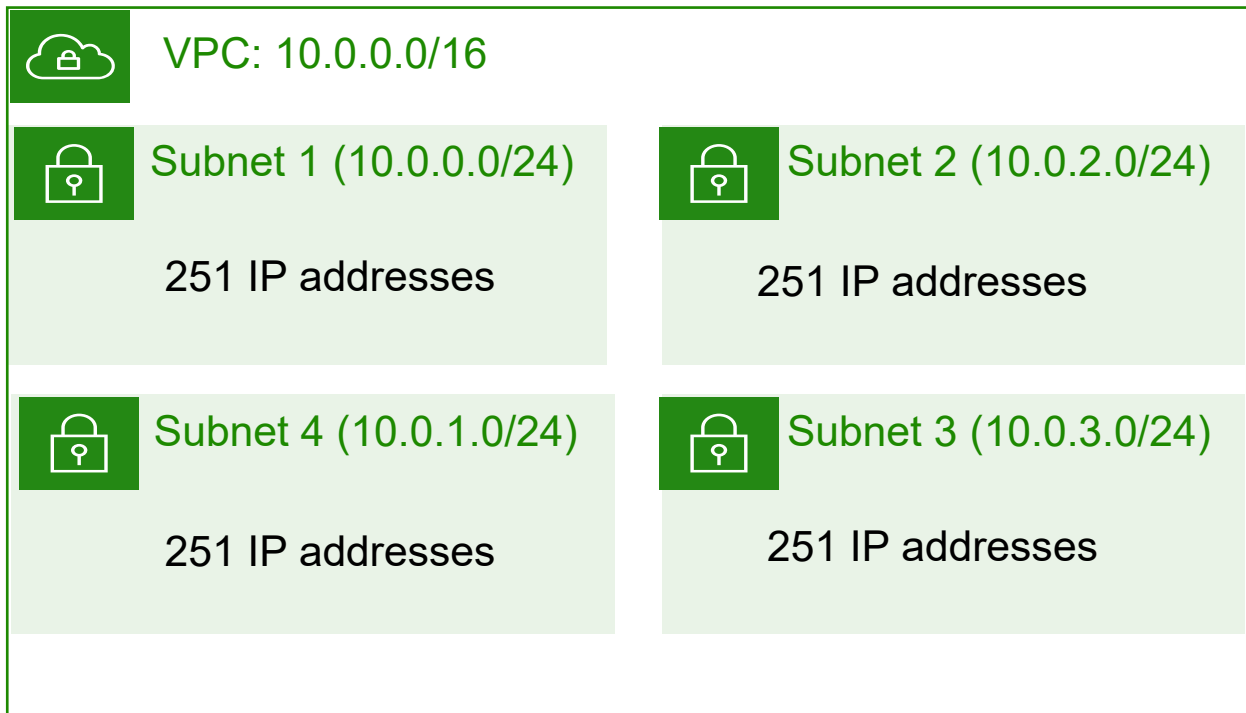
to

x.x.x.x/28 or 16 addresses (min)

4 bits for host address

Reserved IP addresses

Example: A VPC with an IPv4 CIDR block of 10.0.0.0/16 has 65,536 total IP addresses. The VPC has four equal-sized subnets. Only 251 IP addresses are available for use by each subnet.



IP Addresses for CIDR block 10.0.0.0/24	Reserved for
10.0.0.0	Network address
10.0.0.1	Internal communication
10.0.0.2	Domain Name System (DNS) resolution
10.0.0.3	Future use
10.0.0.255	Network broadcast address

Route table

- **Default Traffic Control:** Every VPC automatically comes with a "main route table."
 - This route table acts as the default traffic director for all subnets within the VPC, unless a subnet is explicitly associated with a custom route table.
 - By default, the main route table contains a local route, allowing communication within the VPC itself.
- A VPC can have many other route tables
 - They can be associated with different subnets
- Route table controls traffic at IP level
- Each subnet should only have one route table

Main VPC route table	
Destination	Target
10.0.0.0/16	local

Public and Private Subnet

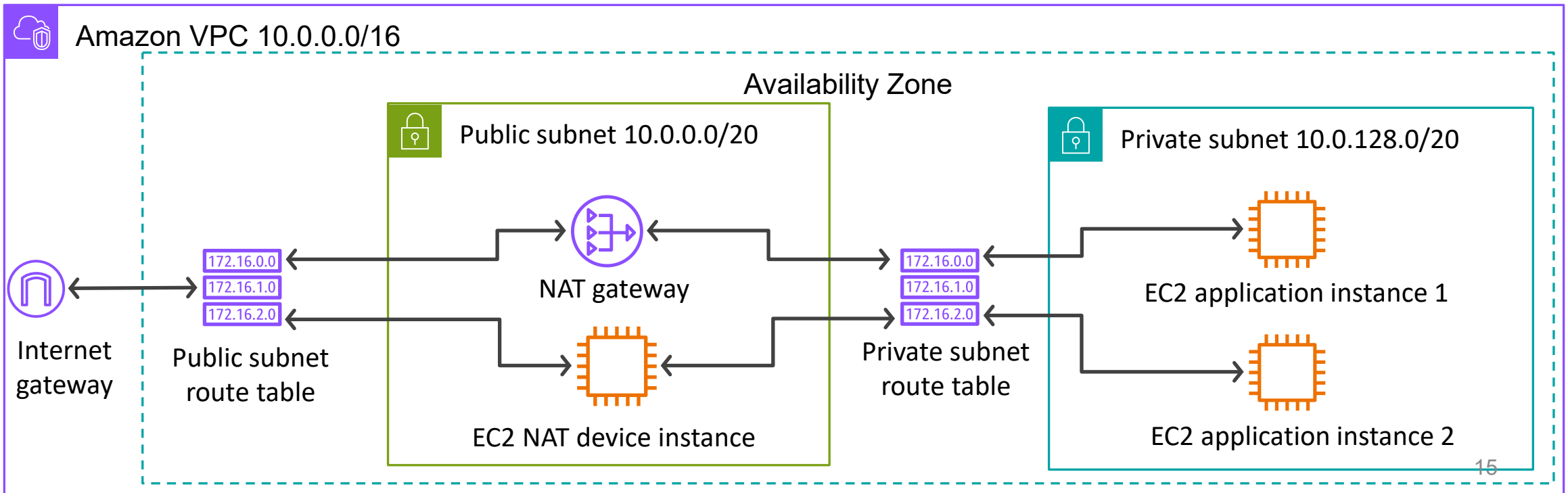
- Public subnet is reachable from the Internet
 - Two way traffic
- Private subnet is not reachable from the Internet, but may access Internet
 - One way traffic
- The accessibility is controlled by customized route tables

Public subnet route table	
Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet gateway ID

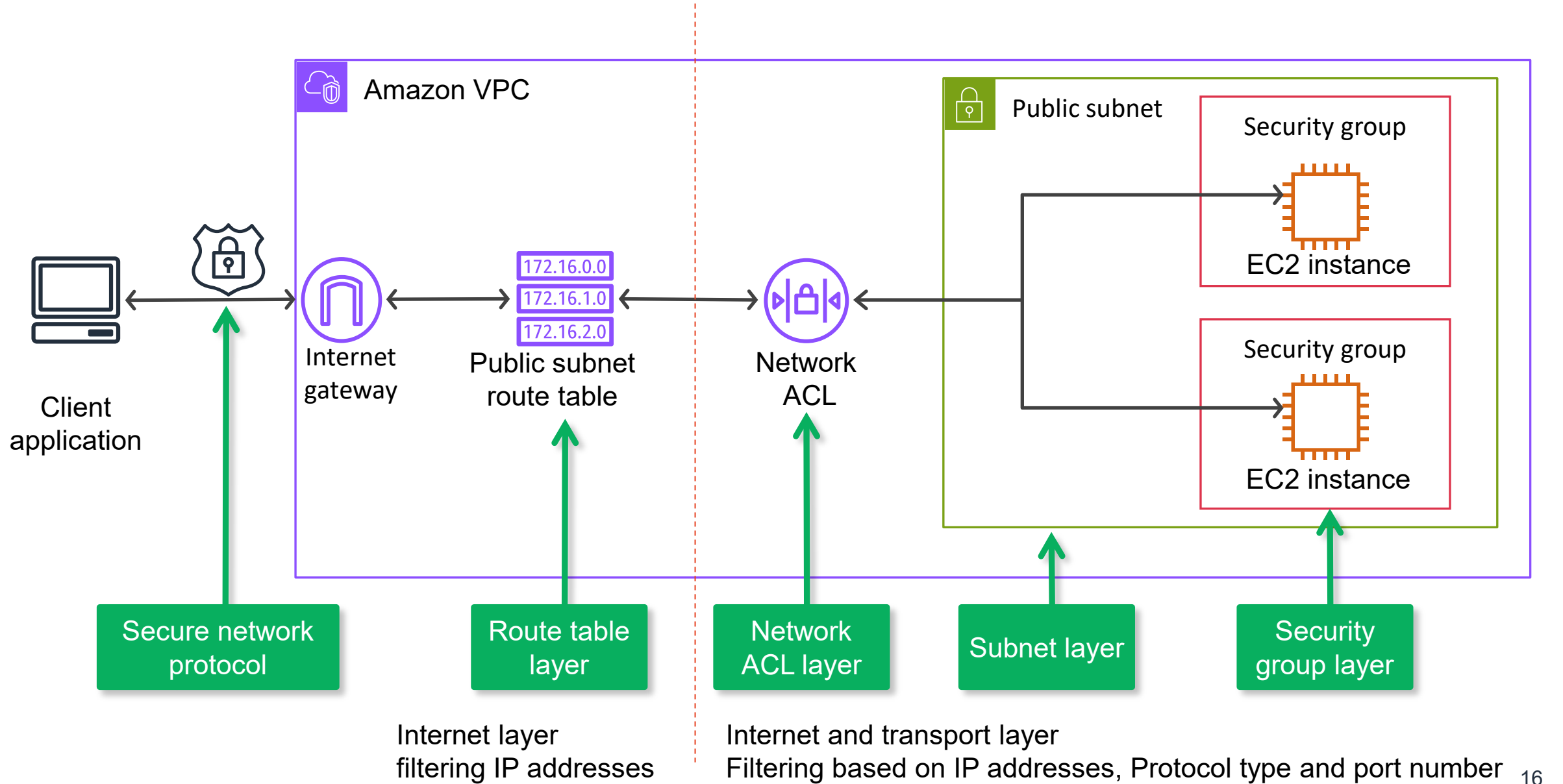
Private subnet route table	
Destination	Target
10.0.0.0/16	local
0.0.0.0/0	NAT gateway ID

Internet Gateway and NAT Gateway

- Internet Gateway is a VPC component
 - There is one per VPC
- NAT Gateway is a subnet component
 - It needs to be placed inside a specific subnet (public subnet)



Security layers of defense



Network Access Control Lists (ACL)

- One Network ACL per subnet
- The Default network ACLs allow all inbound and outbound IPv4 traffic.
- A network ACL has separate inbound and outbound rules, and each rule can either allow or deny traffic.
- ACL rules are stateless
- To enable request-response based traffic, we always need to specify a pair of inbound and outbound rules

Security group

- Security group is a VPC component
- A VPC can have multiple security groups
- Security Groups are primarily applied to **compute resources or services running on compute resources** within the VPC,
 - EC2, RDS, ALB , Lambda
- Security Groups contain **rules that control inbound and outbound traffic** to the associated resources.
- **All Security Group rules are “allow” rules** — there are no explicit deny rules.
- Security Groups are **stateful**: if inbound traffic is allowed, the response outbound traffic is automatically allowed (and vice versa).
- It is common and recommended to specify **other Security Groups as the source or destination** in rules to allow controlled traffic between resources.

Referencing other SG in SG rules

- Security Group references is more flexible than using CIDR range as source/destination
 - Dynamic Membership
 - Simplifies Management
- Typical scenario:
 - **Web Server SG:** Allows inbound HTTP (port 80) from **anywhere** (0.0.0.0/0).
 - **App Server SG:** Allows inbound traffic on port 8080 **only from Web Server SG**.
 - **Database SG:** Allows inbound traffic on port 3306 (MySQL) **only from App Server SG**.

Identity and Access Management

Concept	What it is	Relation to Access
Account user	The original, fully privileged user of the AWS account. Can be authenticated using email/password	Has full access to everything by default; not recommended for daily use.
IAM User	An identity created within AWS account for a person or service. Can be authenticated using email/password or programmatically	Represents an entity that can be granted access via policies
Federated User	A user authenticated externally	Gains temporary credentials after external authentication; Access AWS resources by assuming roles
Group	A collection of IAM users	Helps manage permissions by attaching policies to groups; users inherit those permissions
Role	An IAM identity you can assume temporarily, often by AWS services or users	Used to grant access without long-term credentials; policies attached to roles define what the role can do
Policies	JSON documents defining permissions (allowed/denied actions)	Define what actions identities are allowed or denied to perform

IAM policies and permissions

- **Permissions in AWS** control what actions a user, group, or service can perform on AWS resources.
 - Doing anything with AWS resources requires proper permission
- A policy is a document defines a collection of permissions
- Account user has the permission to do anything to any resources belonging to the account
- All other entities get permission through policy
 - Identity based or resource based
 - There are different ways to specify permissions to allowing one resource to perform action on another resource

S3 permission

- S3 permissions can be specified as resource-based or identity based
- Public bucket means either the bucket policy or the object ACL allows public read access.
 - No further identity-based policy is needed to read from the bucket
- Private bucket may use bucket policy to allow limited access
- We can also use identity-based policy to specify permission for S3 buckets

Allowing EC2 instance to access private S3 bucket

- An IAM role
- A policy attached to the role
- An instance profile
 - An IAM resource created to handle permissions for EC2 instances
 - It manages the temporary credential on behalf of the instance
- EC2 – instance profile – role - policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": "arn:aws:s3:::your-
bucket-name/*"
    }
  ]
}
```

EC2 instance and RDS

- This is an application-level permission managed by RDS itself
- RDS instances manages its users and associated permission
 - EC2 instance need database credential to access the RDS
- Allowing EC2 instance to start/stop RDS is an AWS managed permission

```
{
  "Effect": "Allow",
  "Action": [
    "rds:StartDBInstance",
    "rds:StopDBInstance"
  ],
  "Resource": "arn:aws:rds:region:account-
id:db:your-db-instance-identifier"
}
```

Allowing S3 to publish to SNS topic

- Neither S3 nor SNS can assume role
 - Such permission need to be managed as *resource-based policy*
 - Attache a policy to SNS topic
- ```
{
 "Sid": "AllowS3ToPublish",
 "Effect": "Allow",
 "Principal": {
 "Service": "s3.amazonaws.com"
 },
 "Action": "SNS:Publish",
 "Resource":
 "arn:aws:sns:<your-region>:<your-account-id>:<your-topic-name>",
 "Condition": {
 "ArnLike": {
 "aws:SourceArn": "arn:aws:s3:::<your-bucket-name>"
 }
 }
}
```

# Allow Lambda function to access S3

---

- Lambda function get permission through execution role
- It does not need a separate profile to be associated with the role
- Lambda function – execution role - policy

```
{
 "Effect": "Allow",
 "Action": [
 "s3:GetObject",
 "s3:ListBucket"
],
 "Resource": [
 "arn:aws:s3:::your-bucket-name",
 "arn:aws:s3:::your-bucket-name/*"
]
}
```

# Allowing S3 to invoke Lambda function

---

- We need a resource based policy associated with the lambda function to allow it to be triggered by S3
- When this is done through console, the permission is automatically added

```
{
 "Version": "2012-10-17",
 "Statement": [
 {
 "Effect": "Allow",
 "Action": "lambda:InvokeFunction",
 "Principal": "s3.amazonaws.com",
 "Resource":
"arn:aws:lambda:REGION:ACCOUNT_ID:function:
MyLambdaFunction",
 "Condition": {
 "StringEquals": {
 "aws:SourceArn": "arn:aws:s3:::my-bucket"
 }
 }
 }
]
}
```

# Compute resources

---

- Virtual Machine
  - EC2 powered by Nitro system
- Container
  - Docker or other container run time
- MicroVM
  - Firecracker VM running on bare metal instance to support Lambda function

# Replication and Scalability

---

- Theoretically unlimited copies
- Load balancer to distribute traffic
- Auto scaling to control the number of copies
- Provision unit
  - VM (Auto scaling group)
  - Task of container (ECS)
  - Pod of container (EKS)

# Event Driven Architecture Services

---

- Compute
  - Lambda Function
- Event Router
  - SNS
  - Publisher/Subscriber model
  - Used to fan out messages/events
- Event Store
  - SQS
  - Support point-to-point communication

# Simple Queue Service

---

- The messages are replicated in multiple queue servers to ensure high availability
- Message stays in the queue while they are processed by a consumer
  - Messages being process are hidden from other consumers
  - They will be deleted if the processing is completed within a configurable visibility time out window
  - They will become visible if the consumer does not finish processing within the timeout window or fail to delete it
- Queue types
  - Standard queue
    - At least once delivery
    - Best-effort ordering
    - Message could be processed multiple times
  - FIFO queue
    - FIFO delivery
    - Exactly once processing

# Storage Services

---

- Simple Storage Service (S3)
  - Object based storage
  - Each S3 bucket is created within a specific region
  - The bucket name needs to be globally unique
- Elastic Block Store (EBS)
  - Block storage (like file system storage)
  - Each EBS volume is create within an AZ
  - It can only be attached to EC2 instance within the same AZ
  - It is automatically replicated within the AZ

# Simple Storage Service

---

- Object stored in bucket will be automatically stored in different locations
  - Usually 3 physical copies are stored in different Azs
- Bucket can have version enabled to protect accidental update or delete
  - Each update creates a new version
  - Delete create a special version called delete marker
- Bucket can be replicated
  - Within the same region
  - Across different regions
  - By default, delete markers are not replicated

# Cloud Database

---

- Hosted Database
  - RDS: fully managed by AWS
  - Single AZ, Multi-AZ deployment
  - Read replica configuration
- Cloud Native Database
  - Dynamo
  - Aurora (running as one of the DB engines of RDS)

# Amazon Aurora

---

- Large scale relational database system
- The key innovation is the separation of db engine and storage nodes
  - DB engine runs modified MySQL or PostgreSQL and can have up to 15 readers
  - Data is always replicated 6 times in three different AZs
- It supports very large data volume (64TB as mentioned in the paper)
  - Data are partitioned into fixed sized segments called Protection Groups (PG)
  - There is no indication if the partitioning algorithm take into consideration the actual schema
- The primary DB engine is the one connecting both layers
  - Redo logs are sent from primary to DB readers and storage nodes
- The LSNs are used to ensure completeness at the the storage layer and consistency at the DB layer
  - Multiple important LSNs (VCL, CPL, VDL, etc ) are established and advanced continuously

# Deployment Automation - Infrastructure as Code (IaC)

- Instead of manually creating resources in the AWS Console, CloudFormation allows us to describe infrastructure using a template file.
- Typical structure of a YAML template file
- When a CloudFormation template is deployed, it creates a stack of AWS resources.

AWSTemplateFormatVersion: '2010-09-09'

Description: Simple Web Server

Parameters: User provided value

    InstanceType:

        Type: String

        Default: t3.micro

Resources: AWS resources to create

    WebServer:

        Type: AWS::EC2::Instance

        Properties:

            InstanceType: !Ref InstanceType

            ImageId: ami-xxxxxxx

Use values from other resources or parameters.

Outputs: Value returned after deployment

    InstanceId:

        Value: !Ref WebServer

# Exam Information

# Final Exam Setting

---

- Duration
  - The exam is 2 hours
  - There are extra 10 minutes reading time
- The final exam is closed book; you can bring
  - A physical standard linguistic dictionary
- You will be given scratch paper
- The final exam has 100 points in total
- The final exam is worth 60% of your mark
- The final exam has 40% barrier
- You need to get 40 out of 100 points in the final exam to pass the unit

# Final Exam Script

---

- Question 1 (40 points)
  - 20 MCQ questions, each has a single correct answer
- Question 2 (10 points): IAM identity
- Question 3 (10 points): IAM policy
- Question 4 (6 points): SQS
- Question 5 (16 points): Network (VPC, Subnet) and Security group
- Question 6 (8 points): ASG and ALB
- Question 7 (10 points): Assignment 2 discussion

# Gradescope Exam

---

- Your exam paper will be scanned and delivered digitally to the graders.
- Write your Student Number in the box on the title page of the exam script(scanned using OCR software)



THE UNIVERSITY OF  
**SYDNEY**

|                |  |
|----------------|--|
| Room Number    |  |
| Seat Number    |  |
| Student Number |  |

**ANONYMOUSLY MARKED**

(Please do not write your name on this exam paper)

# Gradescope Exam

- Answer the MCQ questions on the bubble sheet
- Write your SID on the ID box of the bubble sheet
- Each MCQ question has four options

|                |                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | <b>Version</b> <input type="radio"/> A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> E               |
| <b>ID</b>      | <b>Other</b>                                                                                                                                         |
| <b>Section</b> | <b>Marking Instructions</b><br>Be sure to completely fill in the appropriate bubble.                                                                 |
| <b>Date</b>    | <b>Example</b><br><input checked="" type="radio"/> A <input type="radio"/> B <input type="radio"/> C <input type="radio"/> D <input type="radio"/> E |

|   | A                     | B                     | C                     | D                     | E                     |
|---|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 3 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 4 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 5 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

|    | A                     | B                     | C                     | D                     | E                     |
|----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 26 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 27 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 28 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 29 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 30 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

|    | A                     | B                     | C                     | D                     | E                     |
|----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 51 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 52 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 53 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 54 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 55 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

|    | A                     | B                     | C                     | D                     | E                     |
|----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 76 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 77 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 78 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 79 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 80 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

# Gradescope Exam

---

- Answer all other questions on the exam script in the space provided.
- The exam script contains a few extra pages at the end in case you need more spaces
  - Remember to write the question and subquestion number in front of the additional answer.
  - It is also advisable to add a line to indicate that the rest of the answer is on the additional page.

# Exam Techniques

---

- Read each question carefully and make sure you understand it thoroughly before answering it
  - You have 10 minutes to just read the exam paper
  - Plan your time, choose the easiest questions to do first
- If you are uncertain about a question, you should attempt to answer it to the best of your ability. You may include your interpretation, or assumption as part of the answer
  - The exam invigilators are not part of the teaching team, they are not able to clarify anything

# Exam Techniques

---

- Remember to bring your **student ID card**
- You can bring water and snacks
- Please check the university exam info web page for other permitted and not permitted items
- Visit the exam venue during the break to ensure you know how to get there
- Exam venues are usually equipped with many answer booklets, you can request one as scratch paper, but do not write any answer there.
- Remember to answer all questions

# Replacement Exams

---

- <https://www.sydney.edu.au/students/exams/replacement.html>
- “If you have a serious illness, injury or circumstances arise that affect your ability to attend or complete an exam, you may have grounds to apply for special consideration or special arrangements.”
- “If your special consideration or special arrangements application is approved, you may need to sit a replacement exam. **It's important to be aware of how this may affect you in terms of delaying certain aspects of your course.** For example, sitting an exam later **may delay when you receive your results**, which means you are not able to graduate or re-enrol for the next semester until your results are finalised.”
- **Replacement exam period 1** will be held from 7 – 10 July 2026. Replacement exam period 2 will be held between 29 July – 1 August 2025
- **Replacement exam period 2** will be held from 28 – 31 July 2026.

**GOOD LUCK**  
ON YOUR  
**FINAL EXAM!**

YOU'VE  
LEARNED IT.  
NOW SHOW  
WHAT YOU  
KNOW! ☆

CAFFEINE  
+ CODE  
= SUCCESS

CLOUD ARCHITECTURE  
DISTRIBUTED SYSTEMS  
VIRTUALIZATION  
NETWORKING  
SECURITY & COMPLIANCE

You've  
Got  
This! ♡

FOCUS.  
REVIEW.  
SUCCEED.  
REPEAT.  
♡

CLOUD  
POWERED

#### PLAN FOR SUCCESS

- ✓ Review Notes
- ✓ Practice Questions
- ✓ Understand Concepts
- ✓ Stay Calm
- ✓ Give My Best! 😊

One Step  
at a Time  
You'll Get  
There! ☁

Progress  
Not  
Perfection  
♡

FOCUS MODE  
**ON** ☁

Stay Calm  
You've  
Prepared  
Well! 😊

ONE STEP  
AT A TIME  
☁

BRAIN FOOD

THE SKY  
IS NOT THE LIMIT—  
KEEP REACHING HIGH!



BELIEVE IN  
YOURSELF!  
You've got  
this! ♡

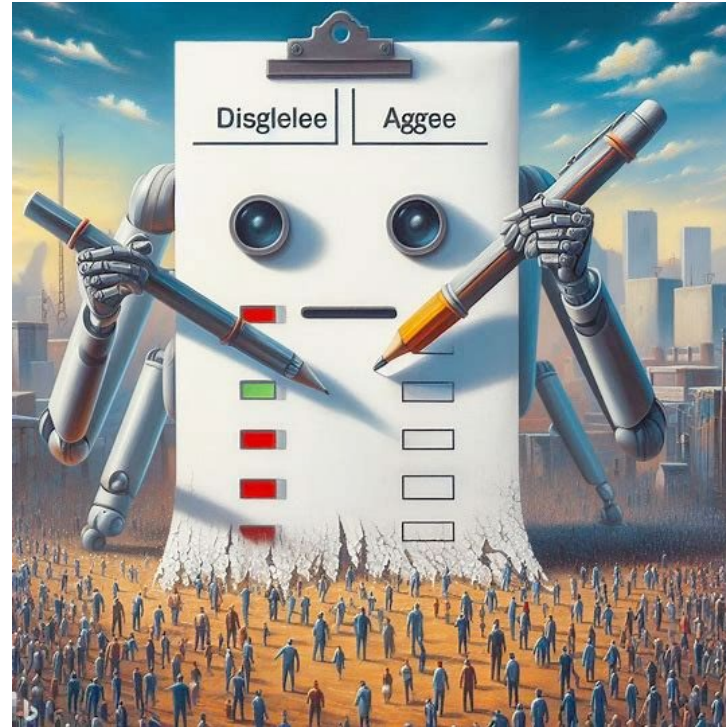
STAY POSITIVE!  
STAY CONFIDENT!  
DO YOUR BEST! ☆

GOOD LUCK AND ALL THE BEST IN YOUR FINAL EXAM! ♡

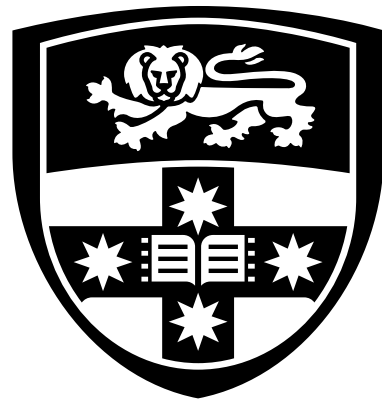
# USS Survey Reminder

---

- <https://student-surveys.sydney.edu.au/students/>



powered by **DALL·E 3 2023**



THE UNIVERSITY OF  
**SYDNEY**